Seara Data

Seara, Menemani Perjalanan
Datamu Lebih Terarah.

Portfolio Bootcamp Data Analyst – Power BI

Batch 2

By Rahmanida S

Bootcamp Data Analyst

Tools Wajib Data Analyst

Excel

Power BI

Python

SQL

4 Weeks Intensive Bootcamp

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Datamu Lebih Terarah.

About Me

As a Chemistry graduate and an experienced QA/QC Supervisor with a four-year track record in manufacturing, I focus on driving process optimization and problem-solving. In today's landscape, I observe that the most impactful operational decisions are entirely data-driven. This insight inspired me to develop expertise in data analytics. To leverage this expertise and expand my career reach toward strategic analytical roles, I am committed to building new technical capabilities through this Data Analyst Bootcamp.

Positioning myself for a strategic transition into data analytics, I am continuously building technical skills in advanced data processing tools. I aim to merge my industrial experience with these new frameworks to architect smarter, high-impact business decisions.

Data Analyst Learning Journey

Excel

Understand & Prepare Data
Become a Data Analyst
Microsoft Excel Basic
Microsoft Excel Intermediate
Power Query

Output: Clean dataset ready for analysis.

Power BI

Analyze & Visualize Data
Microsoft Power BI Basic
DAX & Data Calculation
Microsoft Power BI Advanced
Visualization & Interactive Report

Output: Interactive dashboard and business insights.

Python

Transform & Automate Data
Python Basic
Data Cleaning & Formatting
Python Advanced
Data Transformation

Output: Processed data pipeline and efficient workflows.

SQL

Query & Extract Insights
SQL Basic
Query & Filter Data
SQL Intermediate
Join, Aggregation & Subquery

Output: Insights directly from databases.

Summary of Power BI Session

What I Learned

1. Extract Data
- Import raw dataset into Power BI Desktop

2. Transform Data
- Clean rows, duplicates, format dates, and create group-by summaries using **Table View**

3. Load Data
- Move clean data into the internal data model memory

4. Data Modeling & DAX
- Connect tables (Star Schema) and write formulas (SUM, DIVIDE, DISTINCTCOUNT).

5. Visualization
- Create interactive dashboards

Summary of PowerBI Session

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Extract Data

Navigator

Display Options

Example + Hands On D3.xlsx [2]

- [x] Dataset
- [x] lookup

lookup

Preview downloaded on Monday, August 10, 2026

SND	Region	Column3	Column4	Product Name	Product C
SND01	Jawa Barat	null	null	Tab Valas	
SND02	Sumatera Selatan	null	null	Special Hajj	
SND03	Jawa Tengah	null	null	Wadiah	
SND04	Jawa Timur	null	null	Tab Hajj	
SND05	Kalimantan	null	null	Tab Syariah	
SND06	Sumatera Utara	null	null	Tab Umroh	
SND07	DKI Jakarta	null	null	null	
SND08	Papua	null	null	null	
SND09	Bali	null	null	null	
SND10	Maluku	null	null	null	
SND11	Sulawesi Utara	null	null	null	
SND12	Sulawesi Tenggara	null	null	null	

Load | Transform Data | Cancel

Extract Data Steps:

- **Select Tables** : Choose Dataset and lookup.
- **Transform** : Clicked to clean data in Power Query.
- **Load:** Skipped because raw data needs cleaning.

Summary of PowerBI Session

Transform Data

1. Format Dates

Jenis Kelamin	Date_Cif_Open
Perempuan	5/21/2021
Laki-laki	5/5/2025
Perempuan	7/19/2022

Jenis Kelamin	Date_Cif_Open
Perempuan	May
Laki-laki	May
Perempuan	July

Problem:

Date formats (Date_Cif_Open) are displayed as full numbers, making it difficult to analyze monthly trends.

Solution:
Transformed dates into text-based month names (May, July) using the Power Query formula Date.MonthName to create a standard summary.

Summary of PowerBI Session

Transform Data

2. Blank Data

	CIF		Cust_Name		Usia
Valid	98%	Valid	98%	Valid	98%
Error	0%	Error	0%	Error	0%
Empty	2%	Empty	2%	Empty	2%

	CIF		Cust_Name		Usia
Valid	100%	Valid	100%	Valid	100%
Error	0%	Error	0%	Error	0%
Empty	0%	Empty	0%	Empty	0%

Problem:

There is 2% empty data in the CIF, Cust_Name, and Usia columns which can reduce data accuracy.

Solution:

Filter out the empty rows using Power Query so that all columns become 100% valid with 0% empty data.

Summary of PowerBI Session

Transform Data

3. Duplicates

CIF	Cust_Name
2287021	Budi Kusuma
2287021	Budi Kusuma
2287021	Budi Kusuma
8369052	Dewi Nugroho
8369052	Dewi Nugroho
4849083	Eka Wijaya
4849083	Eka Wijaya

Problem:

Duplicate data is found in the CIF column.

CIF	Cust_Name
5573033	Bayu Kusuma
7586399	Bayu Nugroho
2287021	Budi Kusuma
3240019	Budi Sari
5085108	Chandra Nugroho
6257928	Chandra Saputra

Solution:

Remove the duplicate rows in the CIF column using Power Query to ensure all records are unique, reducing the total records from 150 to 120 rows

Summary of PowerBI Session

Transform Data

3. Add Column

Balance Prod A	Balance Prod B
1296036	2917443
1331445	8610335
1411191	5761288
1003910	7150798
7319461	7179407

Problem:

There is a need to calculate the combined total of Balance Prod A and Balance Prod B.

Balance Prod A	Balance Prod B	Total Balance
1296036	2917443	4213479
1331445	8610335	9941780
1411191	5761288	7172479
1003910	7150798	8154708
7319461	7179407	14498868

Solution:

Create a new custom column named Total Balance by adding the values of Balance Prod A and Balance Prod B using Power Query.

Summary of PowerBI Session

Transform Data

Dataset

Date_Acc_Open	Product_Code	Balance Prod A	Balance Prod B	SND	Cust Profiling	Flag	Custom
3/10/2020	712	1296036	2917448	SND05	Grade B	Low	4213479
4/3/2024	712	1331445	8610335	SND05	Grade C	Low	9941780
8/24/2025	712	1411191	5761288	SND12	Grade A	Low	7172479
3/11/2021	712	1003910	7150798	SND07	Grade C	Low	8154708

lookup

SND	Region	Column3	Column4	Product Name	Product Code
SND01	Jawa Barat	null	null	Tab Valas	413
SND02	Sumatera Selatan	null	null	Special Hajj	521
SND03	Jawa Tengah	null	null	Wadiah	620
SND04	Jawa Timur	null	null	Tab Hajj	621
SND05	Kalimantan	null	null	Tab Syariah	712

4. Merge Data

Problem:

The region information is stored in a separate lookup table and needs to be combined with the main dataset based on the SND column.

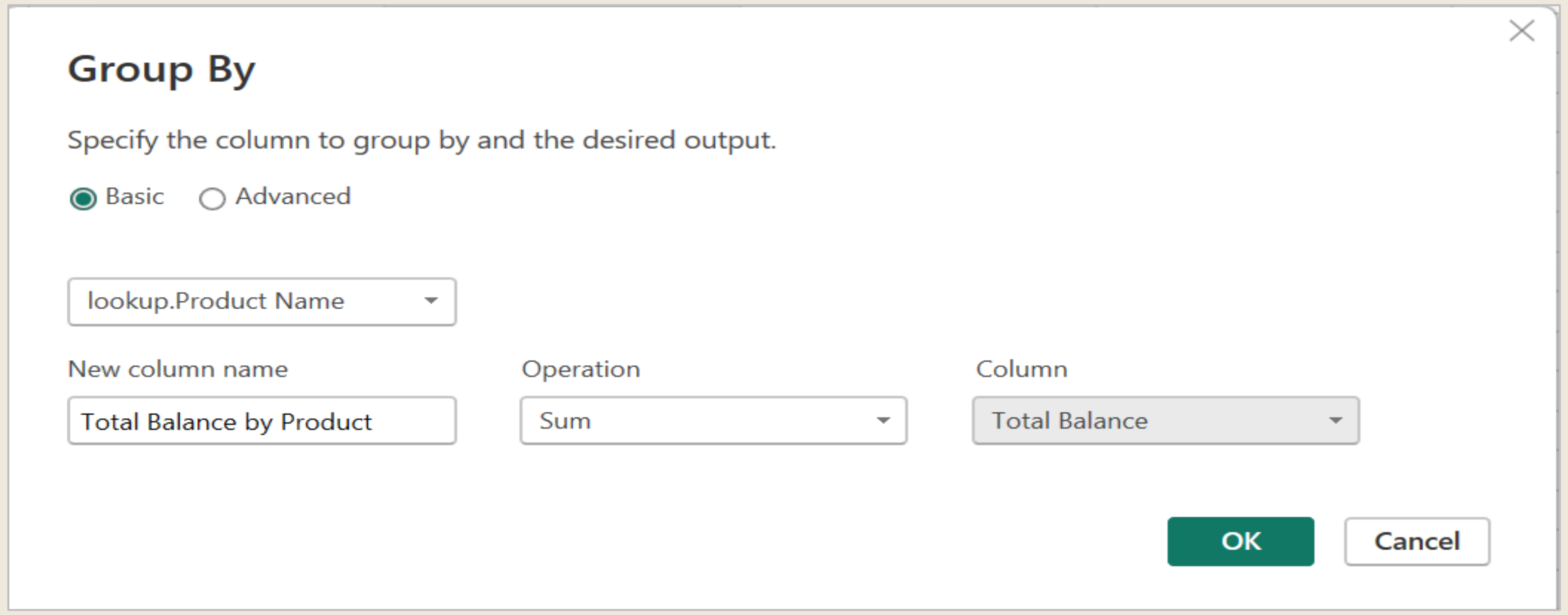
Solution:

Merge the lookup table with the main dataset using Power Query and expand it to bring the Region column into the main table.

Summary of PowerBI Session

Transform Data

5. Group-by



	lookup.Product Name	Total Balance by Product
1	Tab Syariah	1756822984
2	Tab Hajj	848492972
3	Wadiah	1634408451
4	Tab Valas	1083962371
5	Tab Umroh	1191517172
6	Special Hajj	946681077

Problem:
The dataset contains detailed transactions and needs to be summarized to show the total balance for each specific product name.

Solution:
Group the data by lookup.Product Name using the Group By feature in Power Query and calculate the sum to create the Total Balance by Product column.

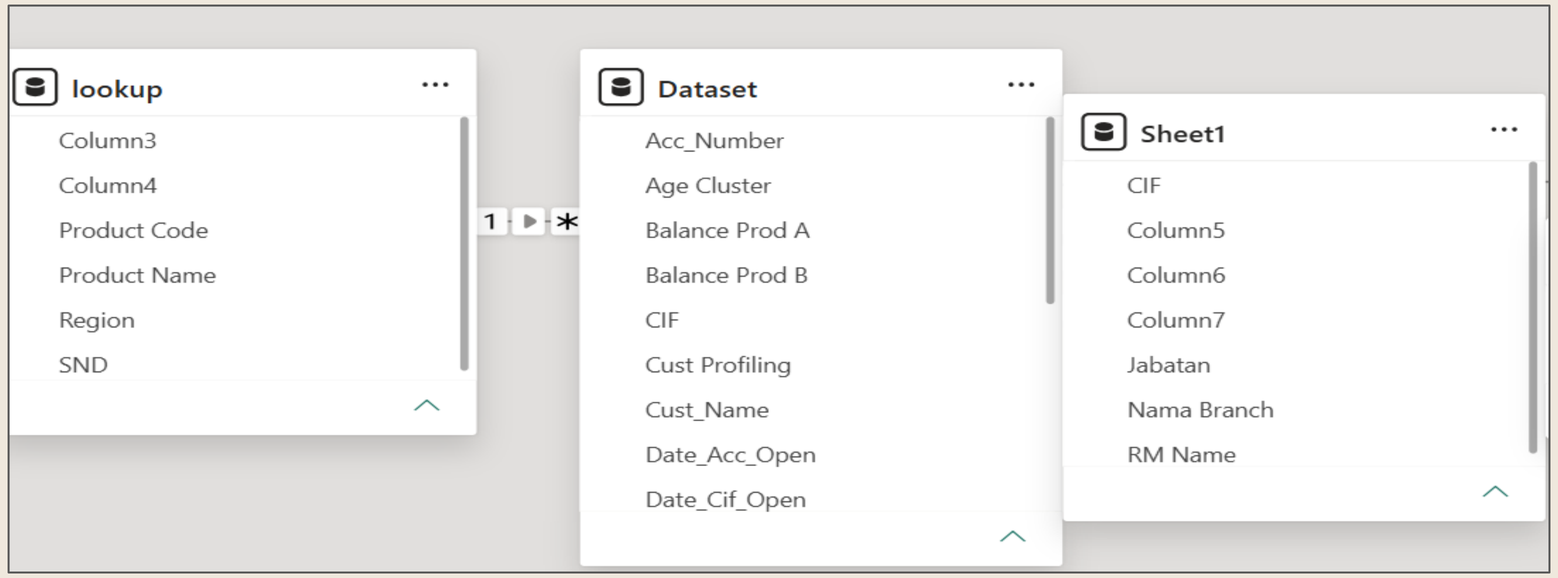
Summary of PowerBI Session

Load & Modeling Data

1. Load Data

If the data cleansing is complete, click 'Close & Apply' to load the data into the data model.

2. Modeling Data



Problem:

The tables in the data model are isolated and need connection to enable cross-table analysis and accurate reporting.

Solution:

Create relationships between the tables using common key columns such as linking lookup and Dataset through the SND column.

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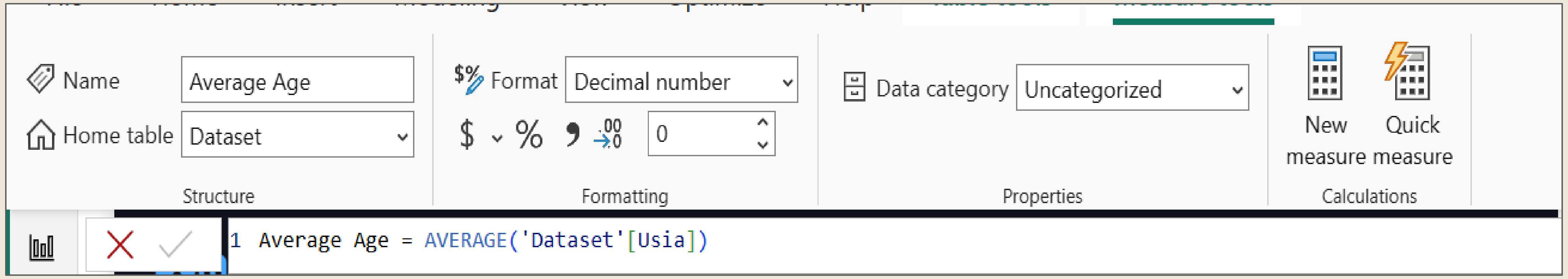
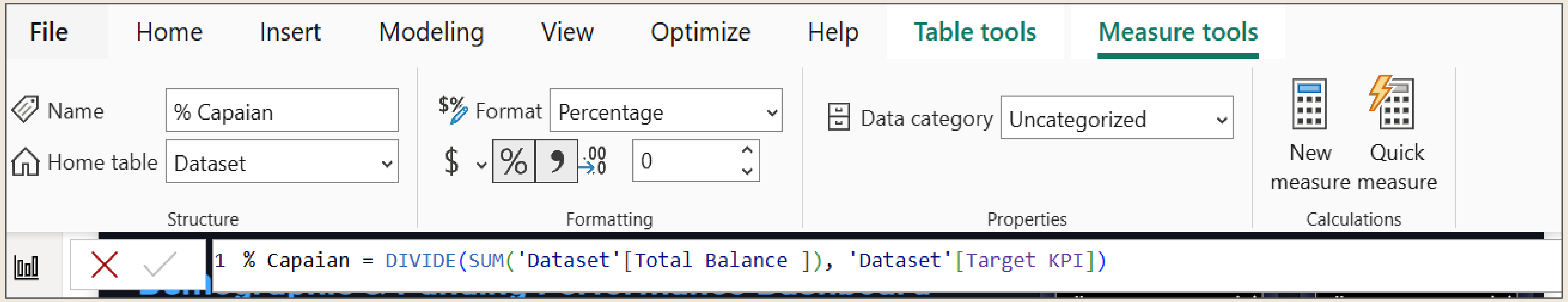
Summary of PowerBI Session

DAX

1. Average & Divide





Create custom measures using the AVERAGE function to find the mean customer age and the DIVIDE function to safely calculate the achievement percentage.

Summary of PowerBI Session

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Visualization

Fitur Used :

- KPI Cards: Show total customers, balance, and average age.
- Gauge Chart: Tracks the current KPI score.
- Line Chart: Shows the monthly total balance trend.
- Bar Charts: Compare balance by region and product.
- Pie & Donut Charts: Show customer segments and profiles.
- Slicers: Filter data by gender, month, region, and RM.

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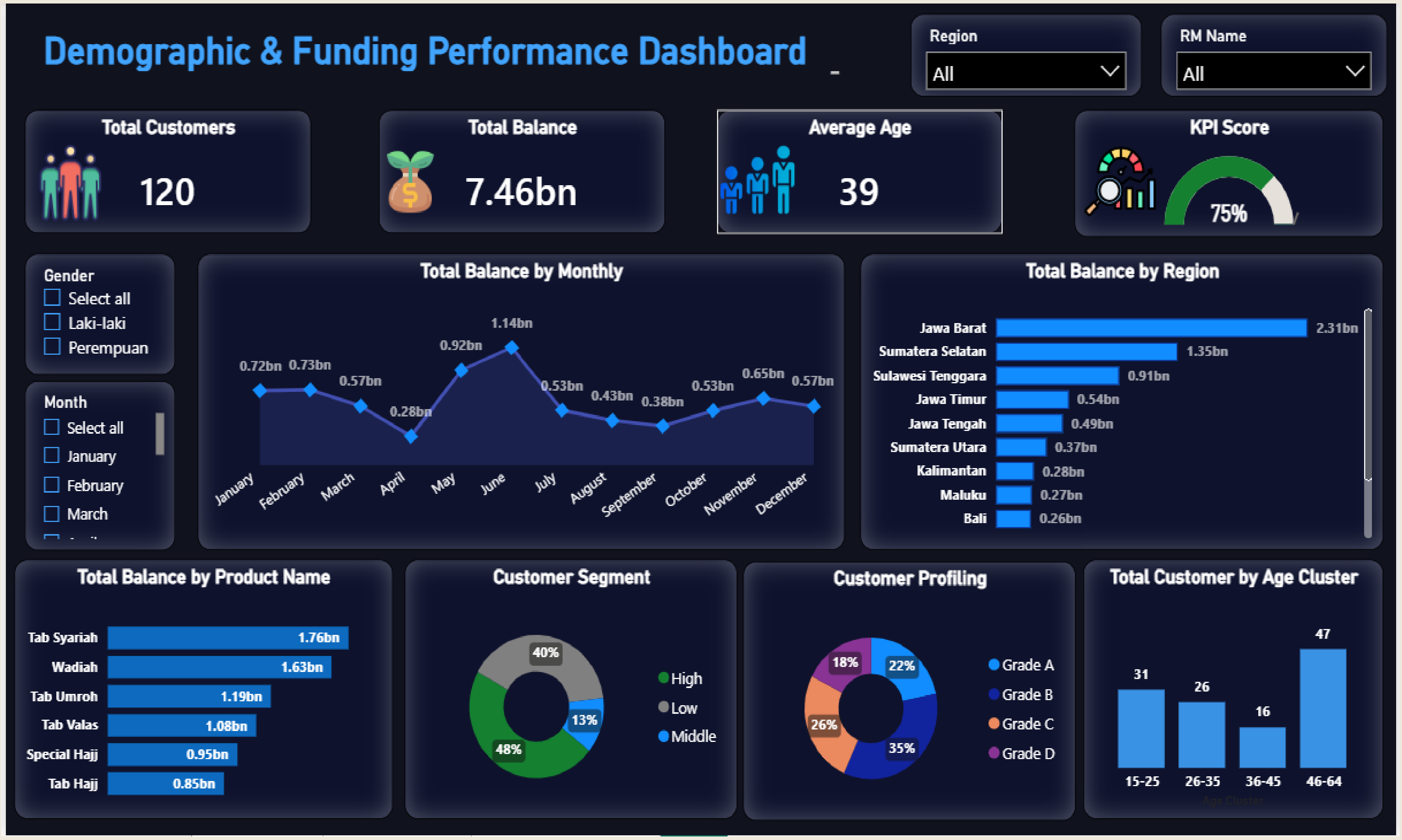
Summary of PowerBI Session

Visualization



Key Insight & Bussines Aggregations :

•**Peak Performance:** Total balance reached its highest peak in June at 1.14bn, while the lowest drop occurred in April at 0.28bn.

•**Top Revenue Drivers:** Jawa Barat is the leading region with 2.31bn in balance, and Tab Syariah is the most preferred product, contributing 1.76bn.

•**Dominant Customer Profile:** The portfolio is heavily backed by the "Low" segment (48%) and "Grade B" customers (35%).

•**Key Age Demographic:** Customers aged 46–64 represent the largest group with 47 individuals, making older adults the primary funding base.

• **Total Portfolio:** Manages a overall funding of 7.46bn across 120 total customers.

•**Operational Efficiency:** Achieved an average customer age of 39 years and a strong KPI score performance of 75%.

Conclusion

Key Takeaways

Through this session, I learned the end-to-end data analyst workflow using Power BI. First, I focused on Extract Data and Transform Data in Table View (Power Query) to clean rows, remove duplicates, fix date formats, and create group-by summaries. Next, I advanced to Load Data to move the clean dataset into the internal memory for Data Modeling & DAX processing, where I connected tables and wrote core formulas like SUM, DIVIDE, and DISTINCTCOUNT. Finally, I performed Visualization in Report View to turn complex raw numbers into an interactive dashboard for clear business insights.

Let’s Connect

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